Tasnia Chowdhury

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Robotics Engineer with 5+ years of experience integrating industrial robotic systems across mechanical, electrical, controls, and software domains.

Columbia University, M.S. in Mechanical Engineering (Robotics), Dec 2020
Drexel University, B.S. in Mechanical Engineering, June 2019

Python | RoboDK | ROS2 | PLC (TwinCAT) | LiDAR, IMU, Embedded Systems | OpenCV | SolidWorks

Ginkgo Bioworks, Robotics Engineer, Emeryville, CA | Apr 2021 - Present

- Integrated and deployed 100+ robotic automation platforms for laboratory manufacturing systems.
- Commissioned 6-axis industrial robotic arms, designing mounting in Solidworks, optimizing motion trajectories in RoboDK simulation and verifying performance on physical hardware.
- Developed acceptance criteria and test protocols for Factory Acceptance Testing (FAT) and Site Acceptance Testing (SAT) to validate deployed robotic systems.
- On site troubleshooting and maintained 100+ robot fleet health utilizing python scripts
- Developed Python drivers to integrate third-party instruments
- Programmed magnetic rail platforms for automated part handling
- Built python tools, reducing system deployment time by 90%
- Deployed PLC control logic using Beckhoff TwinCAT
- Integrated IMU sensor for auto orientation of robotic arm using microcontroller

ARoboticsCompany, Robotics Intern, Ossining, NY, Sept 2020 – Jan 2021

- Designed and configured industrial robotic system electrical wiring and software integration
- Integrated multi-sensor systems (LiDAR, IMU) for real-time robotic perception and control
- Designed mechanical components and prototypes using SolidWorks

Columbia University, Robotics Teaching Assistant, New York, NY, Jun 2020 – Dec 2020

- Mentored students in designing and programming robotic systems using Raspberry Pi, motor controllers, and 3D-printed components.
- Supported development of robotics simulations and computer vision projects using OpenCV and PyTorch

Folia Water, Instrumentation Engineer Intern, Brooklyn, NY, Jul 2020 – Sep 2020

Built automated quality-control rigs using Arduino, fluid pumps, and custom sensor interfaces allowing overnight runs without human present.

Drexel University, Research Assistant, NASA CuPID Satellite, Philadelphia, PA, 2018 – 2019

Simulated heat analysis of satellite components using Thermal Desktop; interpreted and visualized results.

Integrated Project Services, LLC, Project Engineer, Blue Bell, PA, Sept 2017 – Mar 2018

Reviewed component specification sheets, prepared vendor pricing spreadsheets to support contractor selection. Distributed / organized electrical and plumbing drawings